

WACE EXAMS – BONDING QUESTIONS

Section 1: Multiple Choice

2014 WACE EXAM:

4. Which one of the following substances contains only covalent bonds?
- (a) Al_2O_3
 - (b) NH_4NO_3
 - (c) NaCl
 - (d) SiO_2
18. Silicon dioxide does not conduct electricity. Which of the following **best** explains this observation?
- (a) It has delocalised valence electrons.
 - (b) It consists only of non-metal atoms.
 - (c) All of the valence electrons are involved in covalent bonds.
 - (d) It is a covalent network substance.
23. Which of the following solids has the highest electrical conductivity?
- (a) graphite
 - (b) sulfur
 - (c) sodium chloride
 - (d) diamond

2013 WACE EXAM:

8. Which one of the following statements about elements in Groups 1 and 2 on the Periodic Table is correct?
- (a) They can only become positively charged and form strong covalent molecules.
 - (b) They form negative ions because they have few valence electrons.
 - (c) They can either lose or share electrons to form positive ions or a strong metallic lattice.
 - (d) They can form positive ions because they have loosely held valence electrons.

2012 WACE EXAM:

2. A covalent bond would be expected to form between atoms which have the electron configurations
- (a) 2,7 and 2,6.
 - (b) 2,8,8 and 2.
 - (c) 2,8,1 and 2,8,7.
 - (d) 2,8,5 and 2,8,8,1.

4. What would be the chemical formula of the compound formed between magnesium and nitrogen?
- (a) Mg_2N_3
 - (b) Mg_3N_2
 - (c) Mg_2N
 - (d) MgN
6. Which one of the following descriptions relates correctly to an ionic solid?
- (a) an ordered lattice arrangement of cations and free valence electrons
 - (b) a crystalline form that is hard and has lustre
 - (c) a high melting point due to strong intermolecular bonding
 - (d) valence electrons are free to conduct charge
22. Which one of the following compounds contains only ionic bonds?
- (a) CH_3OH
 - (b) HCl
 - (c) NaH
 - (d) NaNO_3
23. Which one of the following statements **best** describes a covalent bond?
- (a) The atoms have a noble gas configuration.
 - (b) The atoms have formed an infinite network.
 - (c) One of the atoms involved in the bond is a metal and the other is a non-metal.
 - (d) There is simultaneous attraction of both nuclei to a shared electron pair.

2011 WACE EXAM:

4. A pure substance is found to be a good conductor of electricity in both its solid and liquid phases. The bonding in the substance is which one of the following?
- (a) ionic
 - (b) covalent molecular
 - (c) metallic
 - (d) covalent network
5. Metals have many characteristic properties, including malleability, conductivity, ductility and high lustre. Which one of the following accounts for these properties?
- (a) the covalent network bonding in metals
 - (b) the mobility of the valence electrons in metals
 - (c) equally-shared electrons between metal atoms
 - (d) covalent molecular bonding between metal atoms

2010 WACE EXAM:

9. A company is designing a new range of saucepans. The base of the saucepan should conduct heat well and evenly while the handle should be a good insulator. Which one of the following lists of substances would be the most appropriate for these tasks?

	Base	Handle
(a)	aluminium	aluminium
(b)	copper	plastic
(c)	glass	glass
(d)	plastic	copper

10. A student has been given a compound that is solid at room temperature and asked to classify it as covalent network, covalent molecular, ionic or metallic substance. Which one of the following sets of observations conclusively identifies the substance as ionic?
- (a) The substance does not conduct electricity as a solid, but does as a liquid.
(b) The substance does not conduct electricity as a solid, but does when dissolved in water.
(c) The substance has a high melting and boiling point.
(d) The substance does not conduct electricity as a solid but is soluble in water.

Multiple Choice Answers:

Year	Question	Answer
2014	4	D
	18	A
	23	A
2013	8	D
2012	2	C
	4	B
	6	C
	22	C
	23	D
2011	4	C
	5	B
2010	9	B
	10	A

Section 2: Written Answers

2014 WACE EXAM:

Question 28

(6 marks)

For the species listed in the table below, draw Lewis structure (electron dot) diagrams.

All valence shell electron pairs should be represented either as $:$ or as $—$

(for example, water $\text{H}:\ddot{\text{O}}:\text{H}$ or $\text{H}-\ddot{\text{O}}-\text{H}$ or $\text{H}-\bar{\text{O}}-\text{H}$)

Species	Lewis structure (electron dot) diagram
CaO	
Cl_2O	

Question 30**(8 marks)**

- (a) Classify the following pure substances into their bonding class by placing each symbol or formula into the appropriate box in the table below. (6 marks)

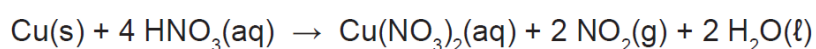
Fe Si NH_4Cl HCl H_2O CaCl_2

Metallic	Ionic	Covalent network	Covalent molecular

- (b) What is it about the bonding of solid metals that explains why they are good conductors of electricity? (2 marks)

Question 31**(12 marks)**

- (c) When concentrated nitric acid is added to copper metal, a brown gas, NO_2 , is produced. The equation is shown below.



0.444 g of Cu reacts with excess 6.00 mol L^{-1} nitric acid.

- (i) Calculate the number of moles of Cu consumed. (2 marks)

- (ii) Calculate the number of moles of NO_2 produced. (2 marks)

Question 28

(4 marks)

For the species listed in the table below, draw electron dot diagrams.

All valence shell electron pairs should be represented either as : or as —

For example, water $\text{H} \begin{array}{c} \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \end{array} \text{O} \begin{array}{c} \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \end{array} \text{H}$ or $\text{H} - \begin{array}{c} \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \end{array} \text{O} - \text{H}$ or $\text{H} - \overset{\cdot\cdot}{\underset{\cdot\cdot}{\text{O}}} - \text{H}$

Species	Electron dot diagram
Na_2S	
SO_2	

Question 31**(10 marks)**

- (a) List **two** features of heterogeneous mixtures that make them different from homogeneous mixtures. (2 marks)

One: _____

Two: _____

- (b) Complete the table with the name of **one** example of each category. (4 marks)

Element	Compound	Heterogeneous mixture	Solution

- (c) In terms of electron transfer, explain how the atoms of magnesium and bromine can react together to form the ionic salt, magnesium bromide. (4 marks)

Question 32

(16 marks)

The physical properties of substances are due to their bonding and structure.

- (a) Complete the following table, outlining the differences in the physical properties of calcium fluoride and fluorine. (6 marks)

	calcium fluoride	fluorine
Type of bonding		
Electrical conductivity as a solid (high, low, nil)		
Electrical conductivity as an aqueous solution (high, low, nil)		

- (b) (i) Does calcium fluoride or fluorine have the higher melting point? Circle your answer. (1 mark)

calcium fluoride

fluorine

- (ii) By describing the bonding present in each solid substance, explain the difference in melting points. (5 marks)

[illegible]

Question 32 (continued)

- (c) (i) Use labelled diagrams to show the bonding structure of solid calcium and solid calcium oxide. (2 marks)

solid calcium	
solid calcium iodide	

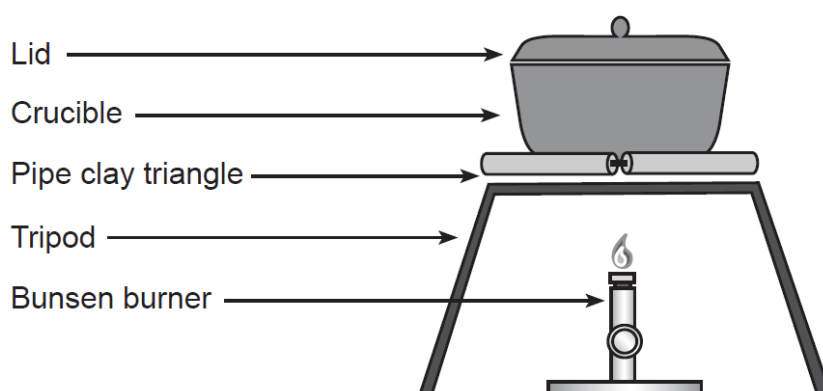
- (ii) Describe how this bonding structure affects each substance's electrical conductivity. (2 marks)

Question 38**(29 marks)**

Some chemists decided to test the hypothesis that magnesium metal combines with oxygen in a 1 to 1 ratio, forming magnesium oxide with the formula MgO .

They used sandpaper to clean a strip of magnesium until it was smooth and shiny all over. They weighed a clean, dry crucible and its lid. Then they placed the strip of magnesium in the crucible, put on the lid and weighed it again.

To get the magnesium metal to react with the oxygen in the air, they heated it over a Bunsen burner flame as shown below. The chemists noticed that when they lifted the lid some white smoke appeared.



- (a) State **two** potential safety hazards of this experiment. (2 marks)

One: _____

Two: _____

- (b) Outline **two** safety procedures which would minimise the risk to the experimenter. (2 marks)

One: _____

Two: _____

- (c) Explain why the magnesium strip was cleaned with sandpaper before it was weighed. (1 mark)

- (d) Explain why the crucible and its contents were heated repeatedly and weighed rather than being heated and weighed just once. (1 mark)

- (e) Give **one** reason why using a lid to cover the crucible while heating was a good idea. (1 mark)

- (f) State **two** reasons for using the pipe clay triangle in this experiment. (2 marks)

One: _____

Two: _____

Question 38 (continued)

After four minutes of heating, the chemists took the crucible and lid off the tripod and weighed it. They repeatedly heated the crucible and its contents and then weighed it every four minutes until the mass stopped changing. They noticed that the shiny silvery strip was now a white powdery ash.

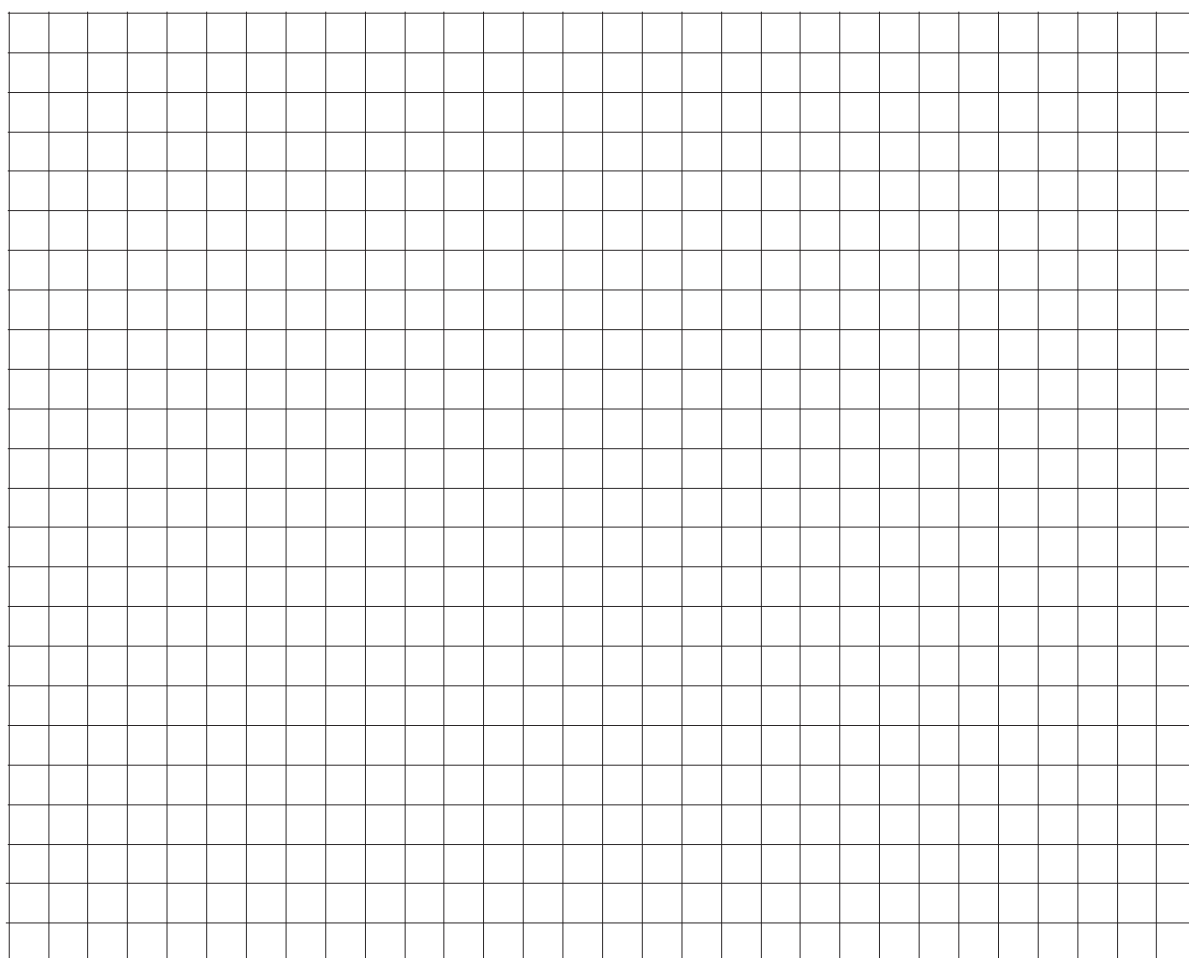
The measurements are shown below.

Mass of clean, dry crucible and lid 21.25 g

Mass of clean, dry crucible and lid plus strip of magnesium before heating 28.75 g

Mass (g) of crucible, lid and strip of magnesium after					
1st heating	2nd heating	3rd heating	4th heating	5th heating	6th heating
30.50	31.70	32.75	33.45	33.65	33.65

- (g) Graph the results using the **all seven** weighings of the crucible and its contents. (4 marks)



A spare grid is provided at the back of this Question/Answer Booklet. If you need to use it, cross out this attempt.

- (h) What was the mass of magnesium before it was heated? _____ (1 mark)

- (i) Calculate the total number of moles of magnesium metal before it was heated. (2 marks)

(j) What was the total number of moles of magnesium present after it was heated? (1 mark)

(k) Calculate the increase in mass of the crucible contents. (1 mark)

(l) Why did the mass of the crucible contents increase? (1 mark)

(m) Calculate the number of moles of oxygen in the magnesium oxide. (2 marks)

(n) What is the simplest calculated ratio between magnesium and oxygen once they have combined to form the compound magnesium oxide? (2 marks)

(o) Based on the experimental results, what conclusion should the experimenters have drawn about the formula of magnesium oxide? (2 marks)

(p) State **two** sources of error in this experiment. (2 marks)

One: _____

Two: _____

(q) Write the balanced equation for the reaction of magnesium and oxygen, showing their states of matter. (2 marks)

Question 28

(4 marks)

For the species listed in the table below, draw electron dot diagrams.

All valence shell electron pairs should be represented either as : or as —

(for example, water $\text{H} \begin{array}{c} \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \end{array} \text{O} \begin{array}{c} \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \end{array} \text{H}$ or $\text{H} - \begin{array}{c} \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \end{array} \text{O} - \text{H}$ or $\text{H} - \overline{\text{O}} - \text{H}$).

Species	Electron dot diagram
CaCl_2	
PCl_3	

Question 32**(9 marks)**

- (a) For each of these two substances, state the type of bonding, a use and the property of each substance that allows it to be used as suggested. (6 marks)

Substance	Type of bonding	Use	Property
graphite			
copper			

- (b) Describe how the theory of bonding in copper can explain the property stated in Part (a). (3 marks)

Question 28

(8 marks)

- (a) For the species listed in the table below, draw electron dot diagrams. (4 marks)

All valence shell electron pairs should be represented either as : or as —

(for example, water $\text{H}:\ddot{\text{O}}:\text{H}$ or $\text{H}-\ddot{\text{O}}-\text{H}$ or $\text{H}-\overline{\text{O}}-\text{H}$)

Species	Electron dot diagram
K_2O	
$NC\ell_3$	

- (b) Diamond and methane contain covalently bonded carbon atoms, yet their melting points are very different (3547°C and -182°C respectively). State the reasons for the difference in melting points in terms of chemical bonding. (4 marks)

[illegible]

Question 28

(4 marks)

For each species listed in the table below, draw the structural formula, representing all valence shell electron pairs either as : or as —.

(for example, water $\text{H}:\ddot{\text{O}}:\text{H}$ or $\text{H}-\ddot{\text{O}}-\text{H}$ or $\text{H}-\overset{\cdot\cdot}{\underset{\cdot\cdot}{\text{O}}}-\text{H}$)

Species	Structural formula
Carbon dioxide (CO ₂)	
Magnesium fluoride (MgF ₂)	

(9 marks)

Explain the properties of each substance in terms of their bonding.

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